



Society of Petroleum Engineers

SPE-227687-MS

Self-Supervised Inversion of Geophysical Data via Physics-Driven Active Learning

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Standard machine learning inversion of geophysical data is difficult to generalize over multiple and diverse datasets. The goal of our research is to enhance machine learning inversion to be robustly applied to field geophysical data. We develop workflows and algorithms that implement a general self-supervised machine learning inversion scheme guided by physics. Self-supervised machine learning inversion is achieved via online training where multiple learning paradigms are combined in an iterative workflow. These consist of active learning, unsupervised learning, reinforcement learning and supervised learning. Physics provides supervisory signals to guide the process. Active learning reduces the data pool while preserving the information content. Unsupervised learning by clustering serves to organize the geophysical data in representative components. Inversion is used as the reinforcement learning agent to generate models proximal to the true models. Supervised learning uses the produced labels for better training. The process converges over a few cycles to enhance and automate geophysical inversion results. The self-supervised machine learning inversion scheme is applied to two field datasets consisting of a helicopter-borne transient electromagnetics (TEM) survey in a desert environment covered by sand dunes, and of a full waveform inversion (FWI) for near surface velocity model building. The helicopter-borne TEM survey is used for mapping the topography of the sand depositional surface and enable accurate statics corrections of the sand body. The self-supervised scheme enhances and de-risk the interpretation of low-relief potential leads. The self-supervised machine learning inversion scheme is further applied for the FWI of a conventional seismic dataset acquired in complex geologic settings. The results provide boosted velocity model building for the near surface while using only 2% of the seismic gathers for training. The developed method shows robustness, accuracy and fast turnaround for multiple geophysical datasets encompassing TEM and seismic FWI. The developed method is a unique self-supervised machine learning inversion method for geophysical applications. The scheme is general such that it can be applied to a large variety of geophysical datasets. The method can be possibly extended to several other disciplines encompassing engineering and science applications where robust, fast and accurate inverse problem solutions are required.

Introduction

Supervised machine learning (ML) for geophysical inversion requires generating many training samples from a broad and unknown parameter distribution. Incorporating physics narrows the parameter space to realistic limits and shifts from big data to "some data/some physics" regimes (Karniadakis et al., 2021), which is ideal for geophysical and engineering applications. In supervised learning, the ML algorithm must learn the data structure to generalize well on unseen samples (test dataset). Similarly to global optimization schemes, like Bayesian approaches, a probability density (PD) distribution must be defined and sampled. For effective ML predictions, the PD used for training should match the PD of the data used for testing. Bayesian schemes, such as Metropolis-Hastings Markov chain Monte Carlo (MCMC) methods, provides the update of the posterior distribution through sampling and evaluations of the likelihood function but are computationally expensive. Physics-driven ML inversion aims to create a cost-effective surrogate process combining physics and ML for efficient parameter space exploration and data space exploitation. Ribeiro and Szepesvári (1996) and Dell'Aversana (2023) propose methods for balancing exploration and exploitation in the inversion process.

Various authors have examined the role of physics in constraining machine learning for addressing the problem of targeting the appropriate parameter space for ML inversion. This exploration includes extensive literature on physics-informed neural networks (PINN) for forward and inverse problem solutions, physics-informed machine learning (PIML) inversion, and self-supervised learning (SSL) for physics-driven inversion. PINNs incorporate Partial Differential Equations (PDEs) and all available information on their solutions to train the neural network to approximate the solutions (Nohra and Dufour, 2024). Embedding PDEs into the loss function of the neural network implements a physics-based regularization by excluding solutions outside acceptable physical bounds (Raissi et al., 2019; Fuks and Tchelepi, 2020). PINNs have been applied in seismic applications (Zhu et al., 2017; Moseley et al., 2018; Waheed et al., 2021; Cao et al., 2022; Cheng and Alkhalifah, 2024), as well as in multi-physics applications such as time-lapse gravity (Celaya et al., 2024) and electromagnetics (Baldan et al., 2023; Nohra and Dufour, 2024). A review of various mathematical approaches for PIML inversion of geophysical data is presented in Schuster et al. (2024).

Recent advancements in machine learning inversion have focused on the field of self-supervised learning (SSL). SSL is a paradigm where a network model is trained on a task using a supervisory signal automatically generated by an auxiliary process that exploits inherent structures and relationships within pools of unlabeled data (Colombo et al., 2025). This differs from the traditional supervised approach, which relies on externally provided labels, such as those from an annotator. Semi-supervised learning (Semi-SL) is a related method that utilizes both labeled and unlabeled datasets. Both SSL and Semi-SL involve a process for labeling a pool of unlabeled data through an auxiliary task generating the supervisory signal. The application of SSL in geophysical inversion is a recent development. Liu et al. (2023) implemented an iterative self-learning mechanism for seismic full waveform inversion (FWI) using multiple approximations on synthetic data. After initial offline training with labeled data, a data misfit loss function is introduced by implementing a forward modeling operation alongside the ML prediction stage. Subsequently, a network-constrained FWI, similar to a standard FWI process, is used to fine-tune the network parameters. One limitation of the semi-SL approach is the need for offline training with manual labeling. Other SSL schemes for FWI are exploring the extension of the data domain to lower frequencies to enable more robust FWI (Hu et al., 2020; Gao et al., 2022).

To reduce computational load in broad parameter spaces, specific strategies are required. Active Learning (AL) from machine learning helps by selecting informative samples rather than random ones (Berger et al., 2023). AL uses heuristics to query the most useful data from a large pool, based on metrics like Euclidean distance and variance (Duda et al., 2000; Settles, 2009). This creates small, effective datasets for faster learning. AL focuses on "diversity" and "uncertainty" for data selection. Applied to geophysical

inverse problems, diversity is implemented via clustering, while uncertainty evaluates misfit functions on geophysical responses (Colombo et al., 2025). The AL mechanism allows for a significant reduction in the unlabeled pool of data while preserving the crucial information required for the learning process. In geophysical applications, the described AL sampling techniques are applied to the geophysical responses, utilizing recorded datasets, such as seismic records or electromagnetic responses, for analyzing underground geological diversity (e.g., by the Euclidean distance) and quantifying uncertainty metrics by sampling data with larger residuals (e.g., through a greedy algorithm). Some advanced schemes also evaluate the probability distribution of the residuals, where various methods (e.g., Markov Chain Monte Carlo - MCMC) can be employed for the sampling process (Wu et al., 2022).

The reduced informative data pool obtained through AL is still unlabeled and needs a labeling mechanism for supervised learning. Labels can be generated by an "oracle" or preferably by an automated process. Di and Abubakar (2022) present an iterative AL-CNN (convolutional neural networks) workflow for seismic image interpretation, using reconstruction errors in CNN predictions to identify sections needing further expert annotation for the next training iteration. Their process still requires a human annotator, where we substitute it with a numerical process.

This review examines the research conducted on related subjects, demonstrated with applications to land seismic full waveform inversion (FWI) and transient electromagnetic (TEM) data (Colombo et al., 2021a, b, 2023, 2024, Colombo 2025). The aim is to showcase the efficiency of geophysical inversion achieved through integrating multiple machine learning paradigms, resulting in novel workflows and specialized physics-driven algorithms. Advantages include faster execution via efficient parameter sampling, improved spatial and vertical resolution, and increased accuracy by thorough exploration of the parameter space. This approach, known as physics-driven deep learning inversion (PhyDLI), is combined with active learning (AL) sampling for optimal performance. Additionally, we demonstrate the maturity and high level of automation attained by this process, enabling robust and efficient application to extensive field datasets without human intervention.

Method

Self-supervised learning is first explained from an epistemic perspective, using core machine learning paradigms to show how machines can acquire knowledge efficiently and autonomously. This abstract understanding is then applied to practical geophysical operations, which involve translation into workflows and specific algorithms for geophysical inversion.

Epistemology of self-supervised learning

Machines can learn autonomously by interacting with the environment, supported by a long-term policy providing penalties or rewards. Actions leading to rewards will have a higher probability of being the correct action after several attempts. This self-supervised learning scheme can be achieved in an efficient way by combining multiple machine learning paradigms in an iterative workflow. Active learning, unsupervised learning, and reinforcement learning perform different tasks that, when combined, assign labels to a reduced and fully informative data pool used for supervised learning. Such concepts of learning and inference are adapted to the case of geophysical inversion, using application-specific field data to guide the learning process.

Geophysical learning begins with an AL query on data diversity. Observed geophysical data, such as seismic traces or transient EM decay curves, are analyzed through unsupervised learning using clustering methods like k-means (Hastie et al., 2009). The diverse classes of observed field data, obtained by the clustering operation, reflect the diversity of the underground geology and provide the means for implementing the "diversity" AL query heuristic for the data pool. The optimal number of clusters is found using an "elbow" curve, which relates error measures (e.g., RMSE) of each cluster member relative to the

corresponding cluster prototype. Multiple instances of k-means clustering allow us to determine the best trade-off between error and number of clusters, which is represented by the elbow of the resulting curve. The identified clusters are then randomly sampled to generate a sufficient statical pool for the successive step of label assignment. Labels (i.e., output model parameters) are generated by an inversion process (i.e., physics-based) that we implement in the following examples as a deterministic (e.g., local optimization) scheme. A forward modeling operation provides data features (i.e., input features) associated with the corresponding inverse models.

The operation described implements the reinforcement learning (RL) paradigm. This can be conceptualized as the action of an intelligent agent interacting with the environment while gaining observations based on a policy that administers penalties or rewards. In the scheme described earlier, the agent is represented by an inversion process (deterministic or stochastic). The policy offering "rewards" aims at reducing long-term data misfits, while "observations" refer to model parameter values or distributions resulting from the inversion process. Using physics as an auxiliary process, the inversion agent generates model parameter distributions that approximate the real world. A forward modeling operation applied to the inverted models provides the associated input features for training. Such pairs of models (output) and responses (input) act as proxies for the true subsurface model parameter distributions and the actual observed data. The generated pairs are effectively used by the machine to learn through training about the subsurface conditions and enhancing its generalization capabilities.

Convergence or long-term data misfit minimization is achieved by using a memory mechanism for the inversion agent, which involves coupling physics-driven inversion with machine learning inversion. The coupling is implemented by introducing a penalty on the common model term (Colombo et al., 2021b,c, 2023, Colombo 2024). This process is recursive and includes data augmentation at every cycle, as shown in Figure 1. The process stops automatically when the learning process is complete, as defined by statistical tests.

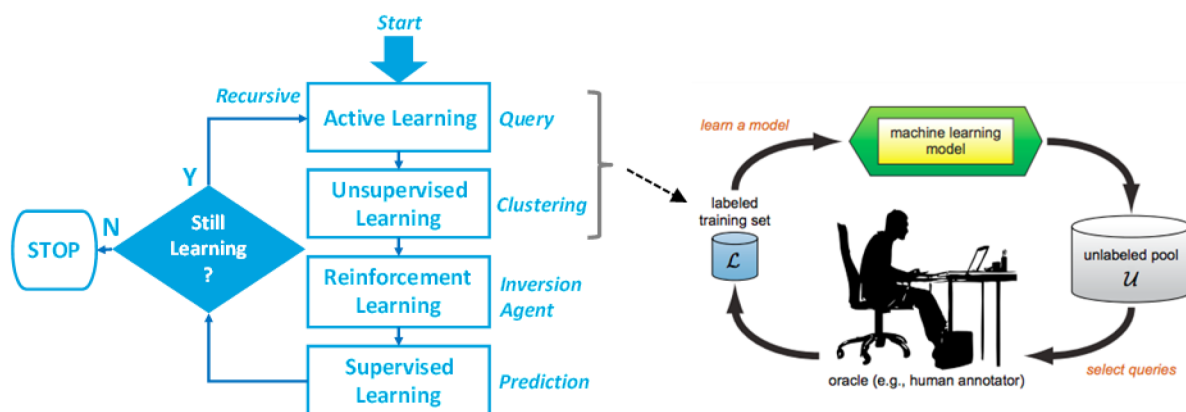


Figure 1—Self-supervised learning for geophysical inversion. Active learning issues a query heuristic for selecting small data pools, unsupervised learning identifies data classes containing relevant information. Reinforcement learning uses an intelligent agent (a physics-based inversion) to interact with the environment (field data) and produce observations (model parameters or labels) under a reward policy (long-term data misfit reduction). Supervised learning utilizes the labeled data for training and predicts model parameters. The process is repeated until the learning process is completed. Graphical design modified from various sources (Colombo et al., 2024; Settles, 2009).

Logical block description

The PhyDLI scheme integrates physics-based and machine learning-based inversion processes by coupling them through penalty functions (Figure 2a). Physics not only bounds predictions but also generates samples aligned with field data, steering the parameter space solution accordingly. The physics-based inverse problem, whether deterministic or stochastic, operates on a reduced subset of input data selected via AL queries. Physics-based inversion and ML optimization are linked by the common model penalty term,

influencing each other at each cycle by generating new penalties, data augmentation, and re-training for ML. This iterative process refines initial conditions, enhancing exploration of the model space and allowing deterministic schemes to approach global solutions even with non-convex objective functions (Figure 2b). We will illustrate these concepts with synthetic examples and apply the insights to field data.

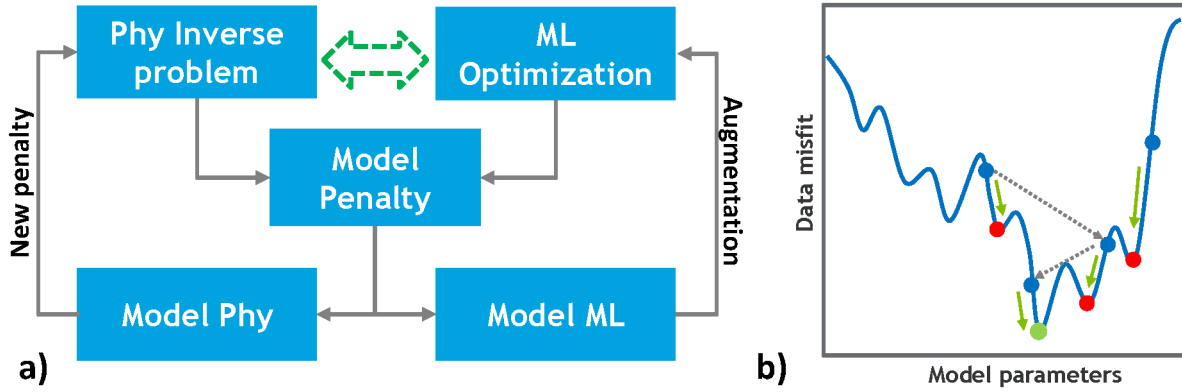


Figure 2—(a) Logical block description of the PhyDLI process; (b) Conceptual reshuffling of initial conditions (dotted arrows) at each cycle, with blue dots indicating the starting points of local optimization processes (solid arrows), red dots for local minima, and green dot as the global minimum.

PhyDLI algorithm

The PhyDLI method is outlined in Colombo et al. (2021b,c; 2023; 2024) with applications to land transient EM, time-domain EM (TDEM), helicopter-borne transient EM (TEM), and near-surface Laplace-Fourier FWI (LF-FWI). The PhyDLI objective function involves a composite structure using Lagrange multipliers, facilitating the integration of physics-based and ML-based optimizations through a penalty on the model term. Following Colombo et al. (2024), the simplest formulation of the objective function can be expressed as:

$$\Phi(\mathbf{m}, \boldsymbol{\theta}) = \Phi_d(\mathbf{m}) + \mu_1 \Phi_{l,m}(\boldsymbol{\theta}) + \mu_2 \Phi_{m,ml}(\mathbf{m}, \boldsymbol{\theta}), \quad (1)$$

where $\Phi_d(\mathbf{m})$ is the data misfit function, $\Phi_{l,m}(\boldsymbol{\theta})$ is the ML objective function, $\Phi_{m,ml}(\mathbf{m}, \boldsymbol{\theta})$ is a penalty term on the model parameters linking the Phy-based inversion with the ML predictions, μ_1 and μ_2 are weights, $\mathbf{m}$ indicates the model vector of the geophysical properties, and $\boldsymbol{\theta}$ indicates the parameters of the ML scheme (e.g., weights and biases). The Phy-based data objective function is described by:

$$\Phi_d(\mathbf{m}) = \|\mathbf{W}_d(\mathbf{G}\mathbf{m} - \mathbf{d}_{obs})\|_2^2, \quad (2)$$

where $\mathbf{d}_{obs}$ indicates the observed data, $\mathbf{G}$ is the Jacobian matrix, and $\mathbf{W}_d$ is a data weighting matrix.

The ML loss function $\Phi_{l,m}$ is described by:

$$\Phi_{l,m}(\boldsymbol{\theta}) = \|\mathbf{H}_0^\dagger \mathbf{d}_t - \mathbf{m}_t\|_2^2, \quad (3)$$

where $\mathbf{H}_0^\dagger$ is the pseudoinverse operator parameterized by $\boldsymbol{\theta}$, and $\mathbf{m}_t$, $\mathbf{d}_t$ are the model parameters and the associated forward response, respectively, of the training dataset. The penalty term $\Phi_{m,ml}$ is used to constrain the Phy inversion model $\mathbf{m}$ with the ML output parameter predictions $\mathbf{m}_t$, such that:

$$\Phi_{m,ml}(\mathbf{m}, \boldsymbol{\theta}) = \|\mathbf{W}_m(\mathbf{m} - \mathbf{m}_t(\boldsymbol{\theta}))\|_2^2, \quad (4)$$

where:

$$\mathbf{m}_t = \mathbf{H}_0^\dagger \mathbf{d}_{obs}, \quad (5)$$

with $\mathbf{W}_m$ being the model covariance matrix and with $\mathbf{d}_{obs}$ representing the observed input data. The solution of the total objective function in [equation \(1\)](#) is obtained by alternate minimizations of the coupled Phy and ML objective functions, as expressed by:

$$\mathbf{m}^k = \underset{\mathbf{m}}{\operatorname{argmin}} \Phi_t(\mathbf{m}, \boldsymbol{\theta}^{k-1}) = \Phi_d(\mathbf{m}) + \mu_2 \Phi_{m,ml}(\mathbf{m}, \boldsymbol{\theta}^{k-1}), \quad (6)$$

for the Phy domain, and:

$$\boldsymbol{\theta}^k = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \|\mathbf{H}_0^\dagger \mathbf{d}_t^k - \mathbf{m}_t^k\|_2^2, \quad (7)$$

for the ML domain, with k indicating the cycle of the recursive procedure ([Colombo et al., 2021a](#)).

In the Phy inversion problem, the coupling to the ML model prediction is direct, whereas for the ML side, the coupling occurs indirectly through data augmentation. At each cycle k of the procedure, the labeled pairs produced by the inversion process and by the corresponding forward modeling are used to augment the ML training base. The new training set at cycle $k+1$ is formed by concatenating the current training set with the previous training sets. The objective function described, combined with a penalty on the model term, is similar to a multi-physics joint inversion objective function where the model parameters may be different in nature. The penalty may be implemented through gradient-based coupling operators that constrain the structure or shape of the models rather than the parameter values ([Gallardo and Meju, 2004](#); [Colombo and Rovetta, 2018](#)).

Synthetic tests

We have examined SSL from an epistemic perspective, operational workflow, and core algorithmic content. Several significant claims have been made regarding the capability of directing and adapting the probability distributions used for training to match the field data under analysis, thereby enhancing machine learning generalization. Additionally, it is claimed that this procedure facilitates thorough exploration of parameter space, even in the presence of non-convex objective functions, by circumventing local minima issues. Furthermore, it is demonstrated that the alternate procedure of physics-based inversion combined with machine learning optimization converges effectively. This convergence must be demonstrated through the reduction of the data misfit. Finally, the proposed procedure is advantageous for addressing geophysical inverse problems both in terms of result quality and efficiency by substantially reducing computational demands.

Transient electromagnetics

The initial test involves synthetic TDEM (or TEM) data generated from a Gaussian-type distribution of model parameters such as the number of layers, layer thickness, and layer resistivity, with a mean of 31.6 m and a standard deviation of 2.5 m. This data serves as the "Training" dataset, representing the parameter space utilized for training. Each 1D resistivity model is characterized by interval resistivity and thickness, modeling the first 200 meters of the ground. The second set, defined by a Gaussian distribution with a mean of 5.6 m and a standard deviation of 1.8 m, is labeled as "Alien," representing testing data unknown to the network and characterized by a different PD. [Figure 3](#) illustrates the distinct differences between the Training and Alien distributions and their corresponding data responses.

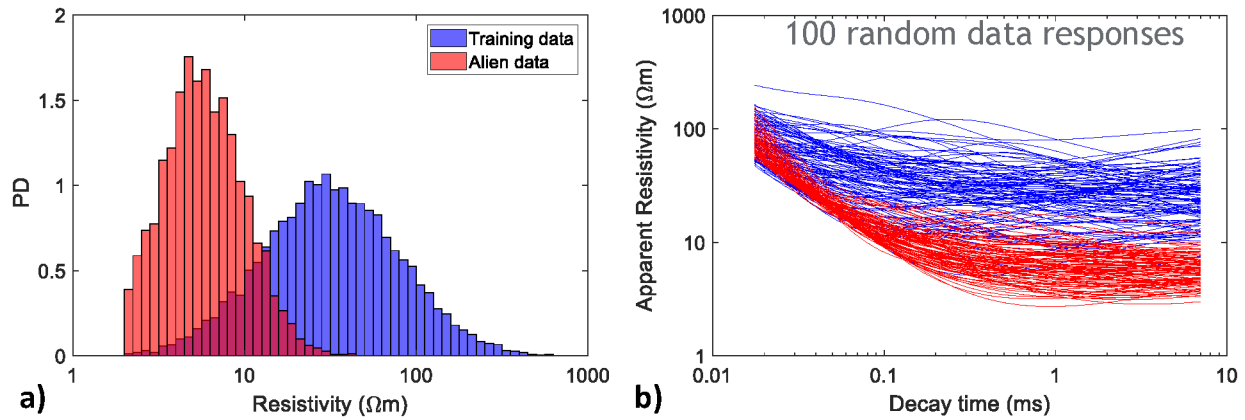


Figure 3—Synthetic TEM dataset comprising "Training" and testing "Alien" datasets:
a) probability density (PD) distributions of resistivity parameters; b) TEM responses.

A shallow fully connected artificial neural network (ANN) with a single hidden layer containing 30 neurons is employed for the test, with different architectures (e.g., U-Net) yielding similar results (refer to Colombo et al., 2021c for details on the training process). The objective is to recover Alien using the PhyDLI scheme, starting from Training as the estimated PD for ANN training.

Figure 4a displays the initial DL prediction of the Alien data, while Figure 4b shows the initial inversion, using a half-space of 10 m as the starting model. Both approaches underperform, with the ML prediction failing to capture the correct distribution, and the Phy-based inversion nearing the solution but heavily biased by the starting model. ML, trained on a different distribution than the testing data, struggles to generalize, while the inversion suffers from an inadequate starting model. By implementing the PhyDLI scheme, the results of both the ANN prediction and deterministic inversion converge towards the diagonal of the parameter cross-plot (Figure 4 c,d). Examination of the PD distributions in the histograms (Figure 4 e,f) reveals that after four cycles of the PhyDLI procedure, the final ANN prediction accurately recovers the true Alien PD. Regarding the inversion, after 40 iterations and beginning from the half-space resistivity model, the results only approximate the true resistivity parameter distribution due to significant influence from the starting model. The final inversion following four cycles of PhyDLI successfully recovers the true PD. Data misfit reduces throughout the cycles, achieving superior results relative to the half-space inversion (RMSE 0.59 for the half-space model versus RMSE 0.17 for PhyDLI). The PhyDLI procedure effectively explores the model space, reduces data misfit, and produces sharper inversion models (Colombo et al., 2021c). The learning process concludes when the data RMSE no longer decreases through the cycles. Additionally, model RMSE is monitored by analyzing the similarity between the ML-predicted model and the inverted model from the inversion process, indicated by the stabilization of the misfit term (Colombo et al., 2021c).

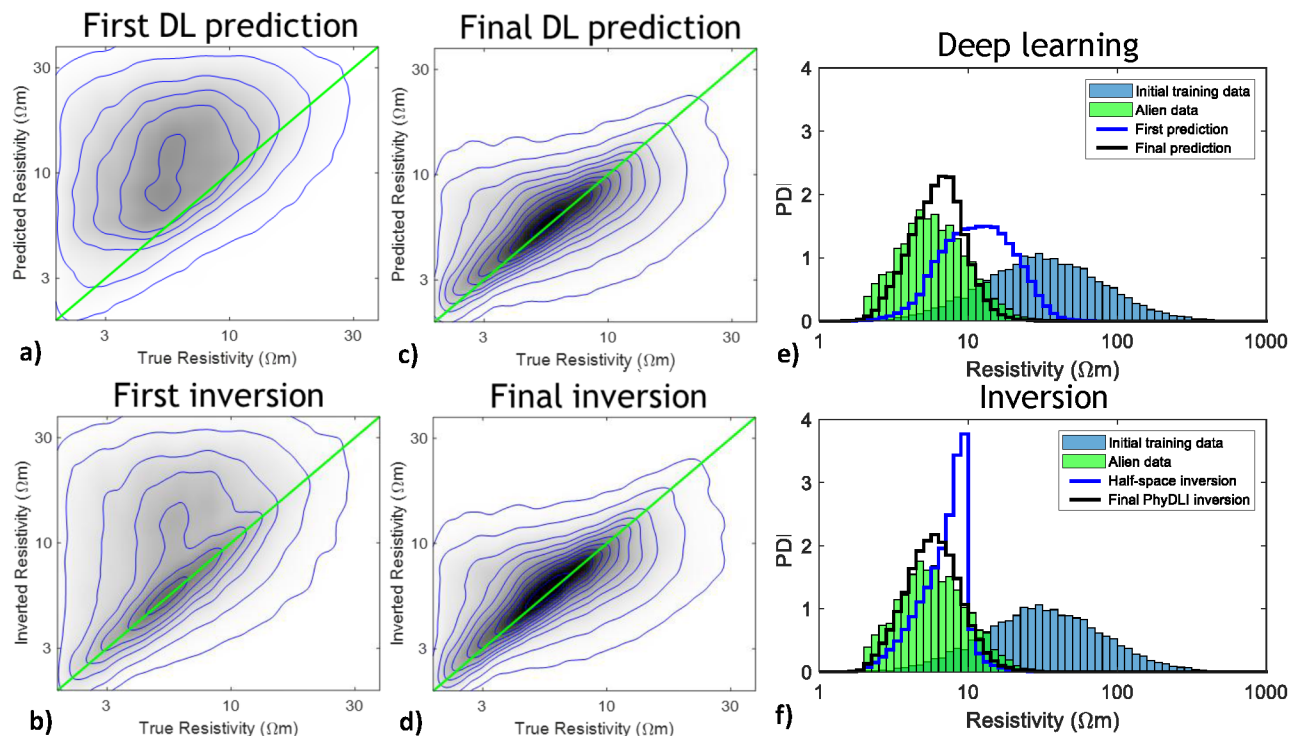


Figure 4—Comparison between PhyDLI and deterministic inversion for the input dataset Alien using cross-plot density functions of true resistivity vs. the predicted/inverted resistivity parameters: a) DL prediction after training with the Training dataset; b) deterministic inversion after 40 iterations using a half-space of 10 #m as the starting model; c) DL prediction after four cycles of the PhyDLI procedure; d) deterministic inversion after four cycles of PhyDLI; e),f) PDs for the different cases.

1.5D Laplace-Fourier FWI

In the first TEM synthetic data test, we evaluated PhyDLI's ability to reconstruct the true Alien PD using an ANN pre-trained with a different PD from the Training data pool. This semi-SL scheme showed that Phy inversion could generate a supervisory signal to guide the distribution toward true values.

In the second synthetic test, we validate the entire procedure, encompassing AL and the associated query heuristics, utilizing the 1.5D Laplace Fourier Full Waveform Inversion (LF-FWI) method, along with model results published by [Colombo et al. \(2021a\)](#). Accordingly, we employ the final 3D FWI velocity model of [Colombo et al. \(2021a\)](#) as the true model we aim to reconstruct and generate the corresponding forward modeled data, which represents our true data in the synthetic test. It is worth noting that the inversion focuses on near-surface velocity reconstruction and adheres to a 1.5D framework, involving a 1D velocity model and a 3D wavefield radiation. The specifics of the 1.5D LF-FWI algorithm are elaborated in [Kontakis et al. \(2023\)](#). A critical element of the 1.5D FWI approach involves the creation of virtual super gathers (VSGs), which comprise high signal-to-noise (S/N) volumetric beams of seismic energy reconstructed within the common midpoint-offset domain ([Colombo et al., 2021a](#)). The generation process for VSGs incorporates transmission-based surface-consistent corrections and beam forming through stacking ([Figure 5](#)). Due to the decoupling from the original acquisition geometry, Machine Learning (ML) applications applied to VSGs yield enhanced generalization. An alternative yet congruent concept of enhancing the generalization capabilities of Convolutional Neural Networks (CNN) via dimensionality reduction through multi-CMP sorting of traces and coincident vertical velocity logs was proposed by [Kazei et al. \(2021\)](#). The VSG beam serves as the input data for inversion and forward modeling in this synthetic test example.

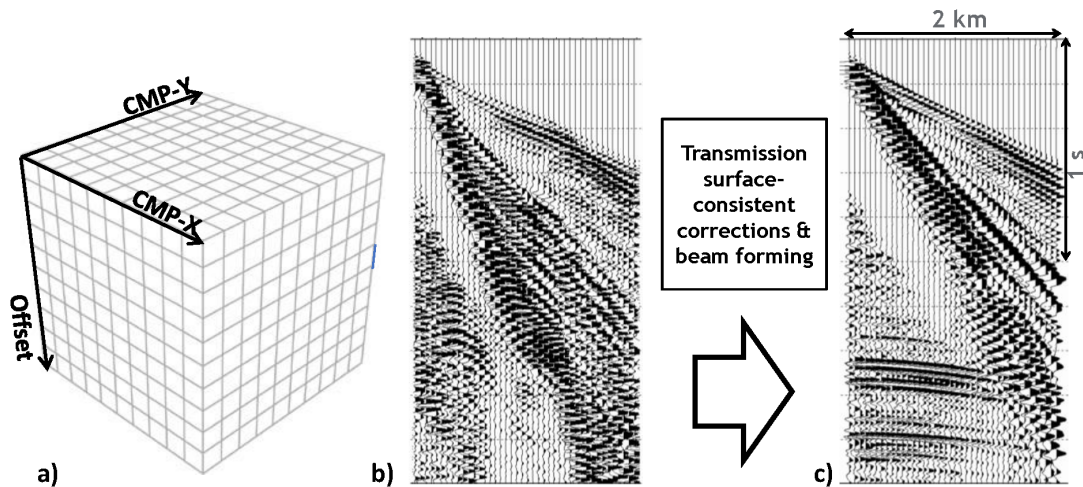


Figure 5—Virtual super gather (VSG) generation process: a) Sorting 3D seismic data in a CMP-offset hypercube (XYO domain) to evaluate surface-consistent decomposition of the transmitted wavefield and derive phase and amplitude corrections (Colombo et al., 2021a), b) Raw seismic record from the SEAM arid model (Oristaglio, 2015), c) VSG coincident with the shot gather position after transmission-based surface-consistent corrections and beam forming through stacking.

The synthetic test's velocity model mirrors a realistic 3D velocity distribution from a FWI process in a complex near surface. It aims to verify the AL-PhyDLI process's capability to reconstruct the model using minimal information, starting with a gradient velocity model. Figure 6 shows cross-sections of the true velocity, AL-PhyDLI reconstruction, and deterministic 1.5D LF-FWI reconstruction. Although results are imperfect, AL-PhyDLI achieves a data RMSE of 3.9% and a model RMSE of 263 m/s, compared to 6.5% and 293 m/s with deterministic 1.5D LF-FWI. The AL sampling data for PhyDLI uses less than 4% of the total VSG dataset, while deterministic FWI uses the full dataset.

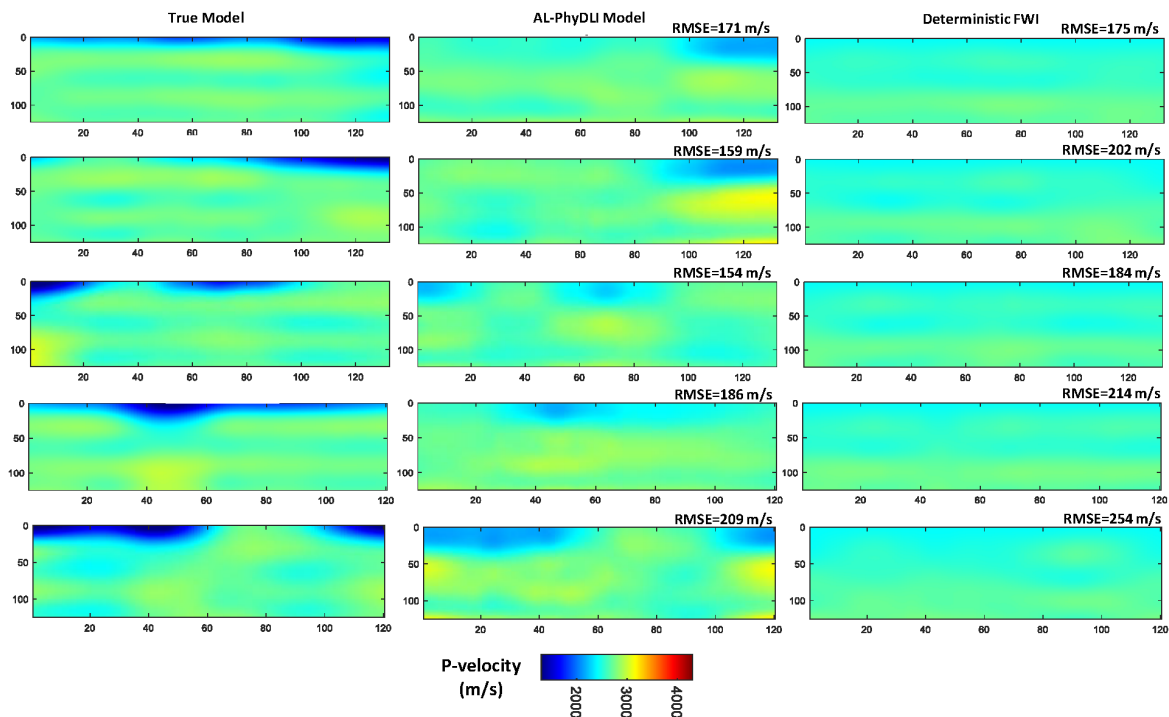


Figure 6—Synthetic test for a 3D velocity distribution: (left) true model from Colombo et al., 2021a; (center) model recovered using AL-PhyDLI; (right) LF-FWI results at 10 Hz, 5s-1, starting from a gradient model.

Figure 7 provides further insights. The AL-PhyDLI process achieves a closer PD distribution of model parameters to the true distribution compared to deterministic FWI (Figure 7a, b). The ML scheme explores the parameter space better, while deterministic FWI is biased by the starting model. Parameter cross-plots (Figure 7c, d, e) show that deterministic FWI fails to align along the diagonal and gets trapped in a local minimum, whereas the AL-PhyDLI process aligns with the diagonal and focuses on the true distribution's lobes. This indicates that the ML procedure can approach the global minimum, unlike the deterministic FWI.

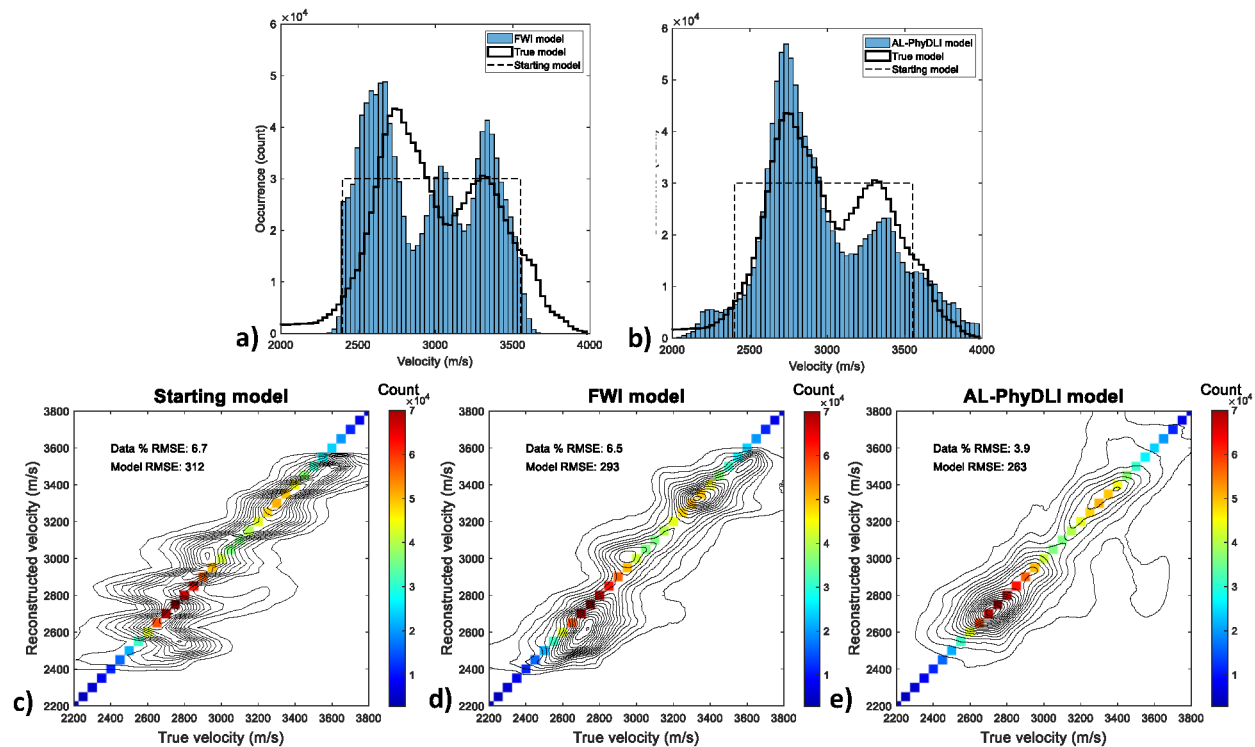


Figure 7—Synthetic test for a realistic 3D velocity distribution: a) PD of deterministic FWI velocity parameters versus true and starting model distributions; b) PD from AL-PhyDLI application; c) cross plot of true vs. starting velocity parameters; d) cross plot of true vs. deterministic FWI parameters; e) cross plot of true vs. AL-PhyDLI recovered parameters (modified from Colombo et al., 2024).

Field data applications

We present two applications of the AL-PhyDLI approach to field data: helicopter-borne TEM in a desert with sand dunes, and velocity model building for complex near-surface conditions using 1.5D LF-FWI.

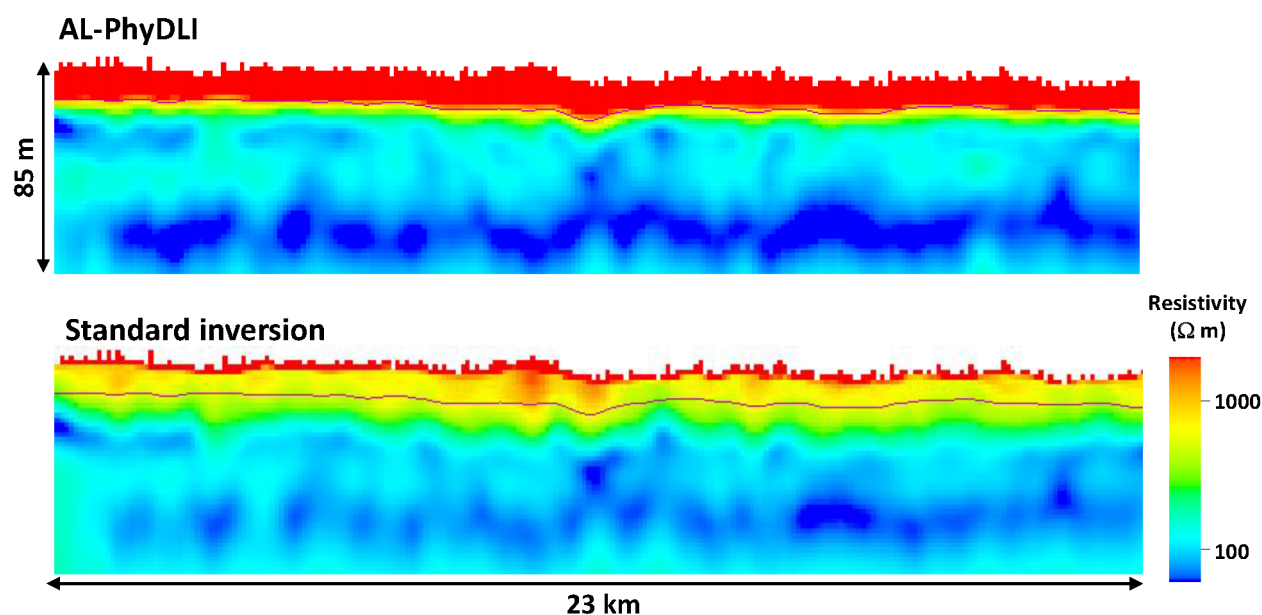
Base of sand via helicopter-borne micro-TEM

The goal is to map the Base of Sand (BoS) accurately for precise seismic corrections of the weathering layer. Incorrect BoS mapping can cause misleading seismic interpretation, affecting low-relief structure exploration. The BoS is a physical boundary for acoustic and electric resistivity, and ultra-resolution, helicopter-borne micro-TEM helps map it accurately (Turkoglu et al., 2018; Colombo et al., 2023). The micro-TEM system has platform-mounted transmitter and receiver coils, each 3 m in diameter and 15 m apart. Suspended 30 m below a helicopter flying at 67 km/h, it provides inline spatial sampling of 2.5 – 5 m after filtering and stacking. With an average terrain clearance of 25 m, it ensures high sensitivity in measurements (Table 1). The system's low transmission moment (45 Am²) and sharp turn-off time allow recording all time windows in the microseconds range (0.35 μ s to 40 μ s), making it ideal for high-resolution shallow mapping with excellent data quality.

Table 1—Summary of acquisition parameters for the micro-TEM survey.

Item	Parameter
Transmitter moment	45 Am ²
Recording windows	0.35 μ s to 40 μ s
Source pulse	Rectangular, 6A, 50 μ s
Measurement frequency	5 kHz
Inline spatial sampling	2.5 – 5 m
Fly line separation	300 m

AL-PhyDLI is used for TEM inversion to precisely define the BoS boundary and utilize the large dataset of 1.2 million micro-TEM soundings efficiently. The ML procedure maps the BoS more accurately than the standard deterministic inversion (Figure 8). AL-PhyDLI employs only 2.5% of field data (around 30,000 soundings) to train an ANN scheme that processes the entire dataset. Similar results are achieved with Gaussian Process regression and a U-Net convolutional neural network using the same training set (Colombo et al., 2023). Figure 9 shows the inferred BoS from standard seismic analysis versus the accurate micro-TEM BoS from ML. A rough paleo topography with a buried drainage channel represent the depositional surface for aeolic sands. Accurate BoS reconstruction significantly alters sand body static correction evaluations and helps de-risking low-relief structures at reservoir depth.

**Figure 8—Comparison between the AL-PhyDLI results and the deterministic TEM resistivity inversion for base of sand.**

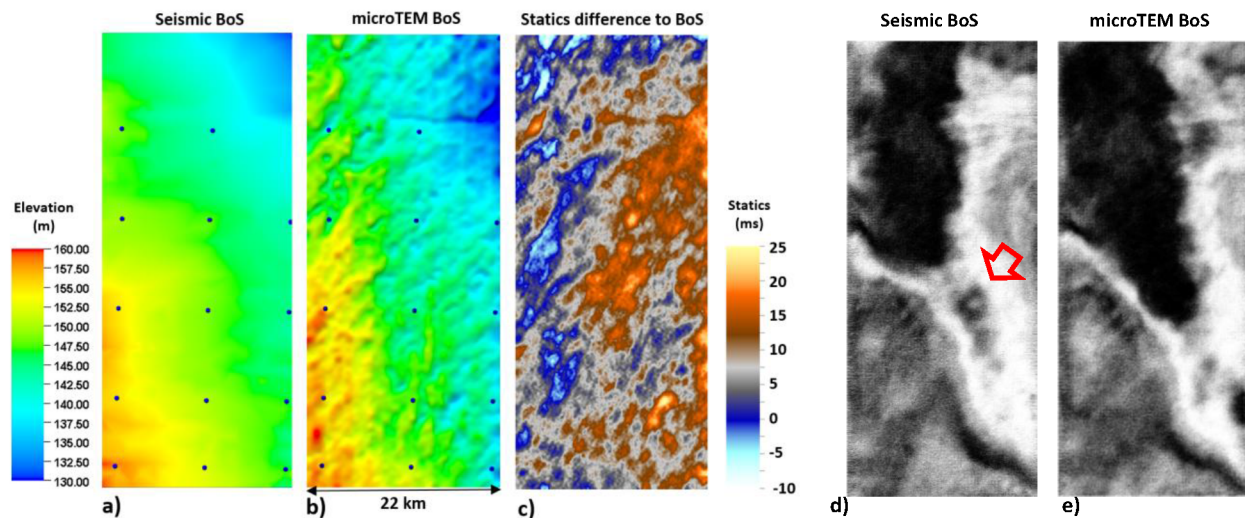


Figure 9—Comparison of sand base determinations: a) seismically derived BoS; b) micro-TEM BoS via AL-PhyDLI procedure; c) difference in vertical time correction for both methods; d) e) reservoir level time stacks showing a likely false low-relief structure (arrow) due to inaccurate sand body corrections.

FWI Velocity model building

A second example of the AL-PhyDLI procedure is shown for near-surface FWI, specifically 1.5D LF-FWI (Colombo 2021a, 2024; Kontakis et al., 2023). This framework approximates near-surface velocity modeling with short offsets or as a starting model for 3D FWI. Using deterministic FWI results from Colombo et al. (2021a) as benchmarks, the same seismic data are used in both cases. The area has complex near-surface conditions due to transcurrent faulting, creating vertical boundaries with flower fault structures and sharp lateral velocity contrasts. Seismic records over 400 km² undergo transmitted wavefield surface-consistent processing, enhancing s/n with beamed VSGs at midpoint positions. The 1.5D LF-FWI generates labels for the PhyDLI process, and AL reduces the final training dataset to 300 samples (less than 2% of available VSG data). AL starts with diversity query through k-means clustering, producing 10 seismic data clusters (Figure 10). Five samples from each cluster form the statistical basis, and GP (Gaussian Process regression) handles the small training dataset efficiently. As shown in Figure 11b, 50 selected VSGs (0.3% of the total dataset) enable sharp shallow velocity reconstruction, outperforming deterministic FWI in resolution compared to a shallow seismic time section.

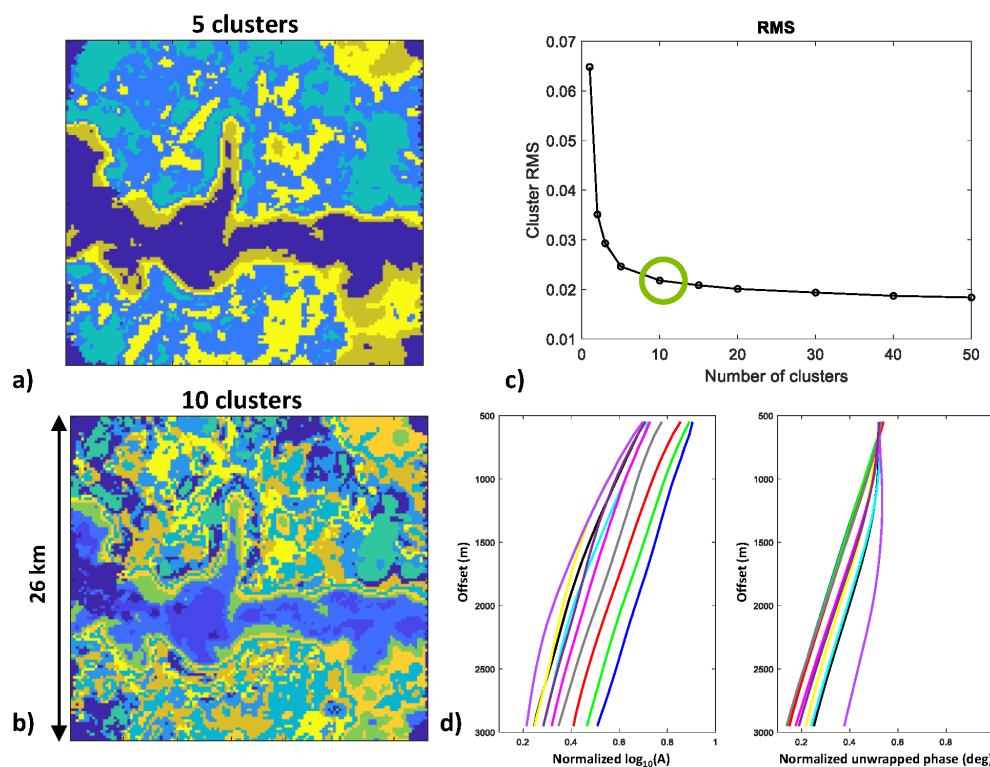


Figure 10—Active learning diversity sampling workflow: a) b) Seismic data organized as VSGs are clustered using k-means for different cluster numbers. c) Optimal cluster number is identified through trade-off analysis at the knee point of the curve. d) Cluster prototypes of the 10 clusters are shown in the Laplace-Fourier domain as normalized log10 magnitude and normalized unwrapped phase for a 10Hz, 5s-1 complex-valued frequency.

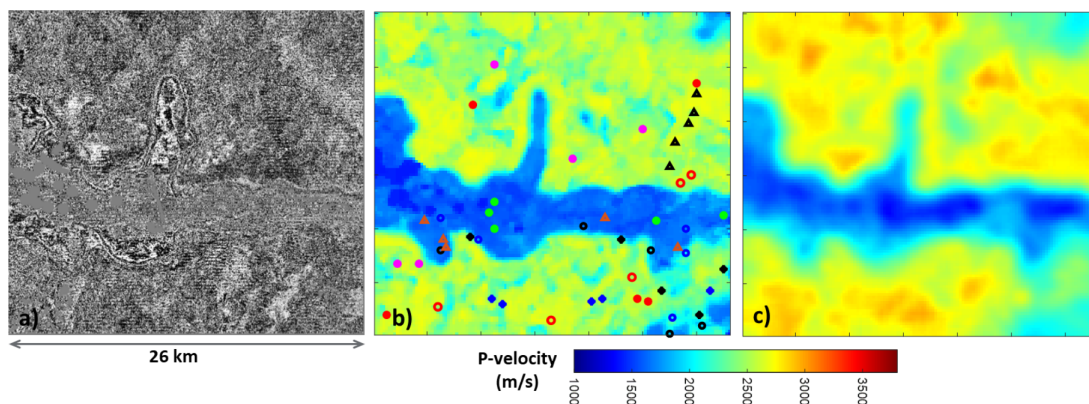


Figure 11—Velocity model building with Active Learning PhyDLI: a) time slice at 164 ms TWT, b) GP velocity prediction at 50m depth after training with 50 pairs of FWI models and forward responses, c) benchmark velocity model at 50m depth from deterministic FWI. Panel b) shows the location of the 50 samples for GP training, with symbols and color coding indicating 10 diverse data types selected via k-means clustering.

In the workflow continuation, the uncertainty query heuristic addresses dataset sections with larger prediction errors. Large data RMSE samples are selected and submitted for further analysis through inversion. The AL-PhyDLI process completes four iterations, reducing the data % RMSE from 7.08% with deterministic FWI to 5.42% with GP prediction, and finally to 4.59% with ANN prediction. A total of 300 samples out of 15,840 (1.9% of the VSG dataset) are used for training. Figure 12 shows the data %MSE distribution sampled for additional inversion and training. The final %MSE distribution indicates that deterministic FWI has the largest error highly influenced by geological features, GP regression yields intermediate results with some geological bias, and ANN results in the lowest error and a randomized distribution with minimal geological bias.

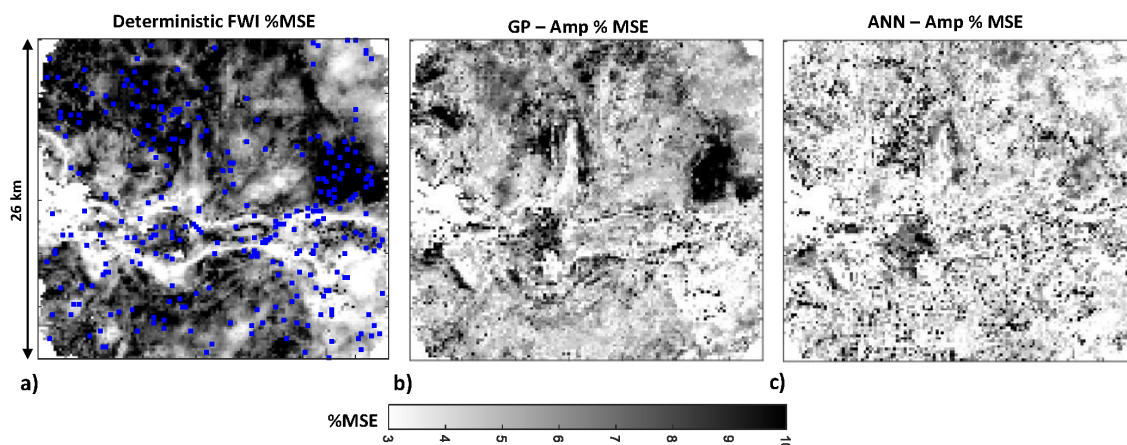


Figure 12—AL uncertainty query heuristic targeting higher %MSE data for further inversion and training: a) benchmark FWI %MSE with selected samples in blue; b) %MSE for the final GP regression model prediction; c) %MSE for the final ANN model prediction.

The results of the velocity model building (Figure 13) demonstrate that the AL-PhyDLI FWI procedure surpasses the deterministic FWI in terms of sharpness and resolution. This conclusion was drawn by comparing it to a benchmark anisotropic pre-stack depth migration, which was obtained independently and calibrated against multiple wells and VSPs.

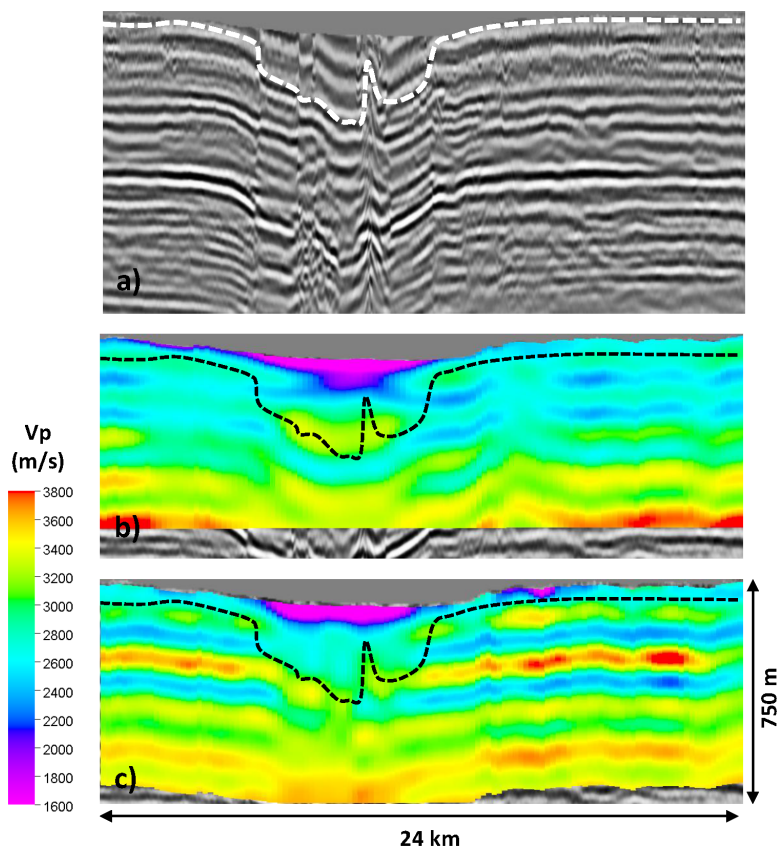


Figure 13—Velocity model results: a) pre-stack depth migrated section through the fault zone; b) velocity cross-section from deterministic LF-FWI; c) velocity cross-section from ANN (AL-PhyDLI process). The shallow horizon helps evaluate the resolution of the velocity reconstruction.

Conclusions

Machine learning offers opportunities to turn high levels of abstraction into practical applications. This contribution demonstrates how conceptualized learning approaches can be converted into effective workflows and algorithms that address complex issues like nonlinear geophysical inversion. The combination of active learning and physics-driven deep learning inversion (AL-PhyDLI) provides a technology for effectively solving challenging non-convex inverse problems for various geophysical data. Computer resource usage is significantly reduced as only a small portion of the data is required to achieve high-resolution inversions. Model details are enhanced due to the improved feature extraction capabilities provided by machine learning. In the discussed self-supervised full waveform inversion application, no offline training occurs; the field data itself guides the learning process, requiring only a few fully informative samples for supervised learning. Active Learning-PhyDLI is a reliable self-supervised scheme suitable for inverting multiple geophysical data types. Given the general formulation of the procedure, AL-PhyDLI could potentially be applied to solving general inverse problems across various scientific fields.

References

- Baldan, M., P. Di Barba and D. A. Lowther, 2023, Physics-Informed Neural Networks for Inverse Electromagnetic Problems: *IEEE Transactions on Magnetics*, **59** (no. 5), 1–5.
- Cao, W., Q. Li, J. Zhang, and W. Zhang, 2022, Accelerating 2D and 3D frequency-domain seismic wave modeling through interpolating frequency-domain wavefields by deep learning: *Geophysics*, **87** (no. 4), T315–T328.
- Celaya, A., B. Denel, Y. Sun, and M. Araya-Polo, 2024, Inversion of Time-Lapse Surface Gravity Data for Monitoring of 3D CO₂ Plumes via Physics Informed Neural Networks: *SIAM Proceedings* PP24, 1–12.
- Cheng, S., and T. Alkhalifah, 2024, Meta-PINN: meta learning for improved neural network wavefield solutions: arXiv:2401.11502v1.
- Colombo, D., and D. Rovetta, 2018, Coupling strategies in multi-parameter geophysical joint inversion: *Geophysical Journal International*, 215.
- Colombo, D., E. Sandoval-Curiel, D. Rovetta, and A. Kontakis, 2021a, Near-surface full-waveform inversion in a transmission surface-consistent scheme: *Geophysics*, **86**, no. 2, U15–U29.
- Colombo, D., E. Turkoglu, W. Li, E. Sandoval-Curiel, and D. Rovetta, 2021b, Physics-driven deep learning inversion with application to transient electromagnetics: *Geophysics*, **86**, no. 3, E209–E224.
- Colombo, D., E. Turkoglu, W. Li, and D. Rovetta, 2021c, Coupled physics deep learning inversion: *Computers and Geosciences*, **157**, 104917.
- Colombo, D., E. Turkoglu, E. Sandoval-Curiel, D. Rovetta, and W. Li, 2023, Machine learning inversion via adaptive learning and statistical sampling: Application to airborne micro-TEM for seismic sand corrections: *Geophysics*, **88**, no. 3, K51–K68.
- Colombo D., E. Turkoglu, E. Sandoval-Curiel, and T. Alyousuf, 2024. Self-supervised, active learning seismic full-waveform inversion: *Geophysics*, **89** (no.2), U31–U52.
- Dell'Aversana, 2023. Inversion of geophysical data supported by Reinforcement Learning: *Bulletin of Geophysics and Oceanography*, **64** (no.1), 45–60.
- Di, H., and A. Abubakar, 2022. Automated active learning in seismic image interpretation: The Leading Edge, Special Section: *Digital transformation*, **41** (no.9), 628–635.
- Duda, R. O., Hart, P. E. and Stork, D. G., 2000. Pattern classification (2nd ed.), Wiley Interscience.
- Farris, S., R. Clapp, and M. Araya-Polo, 2023, Learning-Based Seismic Velocity Inversion with Synthetic and Field Data: *Sensors*, **23**, 8277.
- Fuks, O., and H. A. Tchelepi, 2020, Limitations of physics informed machine learning for nonlinear two-phase transport in porous media: *Journal of Machine Learning for Modeling and Computing*, **1**, 19–37.
- Gao, Z., W. Yang, C. Li, F. Li, Q. Wang, J. Ding, J. Gao, and Z. Xu, 2022, Self-supervised deep learning for nonlinear seismic full waveform inversion: *IEEE Transactions on Geoscience and Remote Sensing*, **60**, 1–18.
- Hastie, T., Tibshirani, R. and Friedman, J., 2009. The elements of statistical learning: Data mining, inference, and prediction (2nd ed.), Springer.
- Hu, W., Y. Jin, X. Wu, and J. Chen, 2020, Physics-guided self-supervised learning for low frequency data prediction in FWI: *90th Annual International Meeting, SEG, Expanded Abstracts*, 875–879.
- Karniadakis, G., I. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, 2021, Physics informed machine learning: *Nature Reviews Physics*, **3**, 422–440.

- Kazei, V., O. Ovcharenko, P. Plotnitskii, D. Peter, X. Zhang, and T. Alkhalifah, 2021, Mapping full seismic waveforms to vertical velocity profiles by deep learning: *Geophysics*, **86** (no. 5), R711–R721.
- Kontakis, A., D. Rovetta, D. Colombo, and E. Sandoval-Curiel, 2023, Efficient 1.5D full waveform inversion in the Laplace-Fourier domain: *Inverse Problems*, **39**, 075012.
- Liu, B., P. Jiang, Q. Wang, Y. Ren, S. Yang, and A. G. Cohn, 2023, Physics-driven self-supervised learning system for seismic velocity inversion: *Geophysics*, **88** (no.2), R145–R161.
- Moseley, B., A. Markham, and T. Nissen-Meyer, 2018, Fast approximate simulation of seismic waves with deep learning: arXiv preprint, doi: [10.48550/arXiv.1807.06873](https://doi.org/10.48550/arXiv.1807.06873).
- Nohra, M., and S. Dufour, 2024, Physics-Informed Neural Networks for the Numerical Modeling of Steady-State and Transient Electromagnetic Problems with Discontinuous Media: arXiv:2406.04380v1.
- Oristaglio, M., 2015, SEAM update: The Leading Edge, **34** (no.4), 466–468.
- Raissi, M., P. Perdikaris, and G. E. Karniadakis, 2019, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations: *Journal of Computational Physics*, **378**, 686–707.
- Ribeiro, C., and C. Szepesvári, 1996, Q-learning combined with spreading: Convergence and results: *Proceedings of the ISRF-IEE International Conference: Intelligent and Cognitive Systems (Neural Networks Symposium)*, 32–36.
- Schuster, G.T., Y. Chen, and S. Feng, 2024, Review of physics-informed machine-learning inversion of geophysical data: *Geophysics*, **89** (no.6), T337–T356.
- Settles, B., 2009. Active Learning Literature Survey, Computer Sciences Technical Report 1648, University of Wisconsin-Madison, 4.
- Turkoglu, E., Colombo, D., McNeice, G., Rovetta, D., Sandoval-Curiel, E. and Yousuf, T., 2018. Imaging sand dunes using helicopter-borne TEM data for seismic near-surface corrections, SEG Technical Program Expanded Abstracts.
- Waheed, U., T. Alkhalifah, E. Haghighat, C. Song, and J. Virieux, 2021, PINNtomo: Seismic tomography using physics-informed neural networks: arXiv:2104.01588.Top of Form
- Wu, C., M. Zhu, Q. Tan, Y. Kartha, and L. Lu, 2022, A comprehensive study of non-adaptive and residual-based adaptive sampling for physics-informed neural networks: arXiv:2207.10289v1.
- Zhu, W., Y. Sheng, and Y. Sun, 2017, Wave-dynamics simulation using deep neural networks, <http://cs231n.stanford.edu/reports/2017/pdfs/542.pdf>.